

The Normal Distribution: μ & σ

A continuous random variable x is said to have a normal distribution with a parameter μ and σ , where μ take value from $-\infty$ to ∞

$$-\infty < \mu < \infty \quad \sigma > 0$$

The pdf of x is defined as,

$$f(x, \mu, \sigma) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

Standard Nor. dis:

(The standard normal distribution with parameter values $\mu=0$ & $\sigma=1$ is called std. normal distribution)

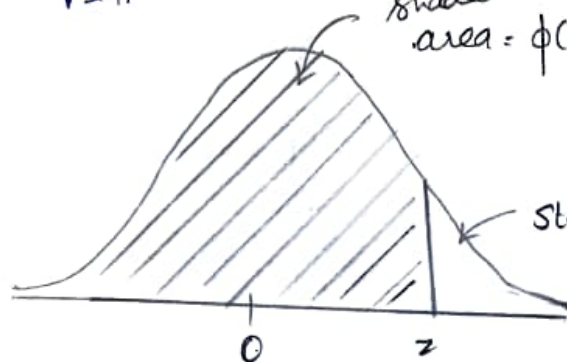
A rv having std. normal distribution is called a std. normal rv & will be denoted by z

The pdf of z is

$$f(z, 0, 1) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$$

$$-\infty < z < \infty$$

shaded area = $\phi(z)$



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ex: 4.13

Determine the standard normal probability

↳ direct table value

1) $P(Z \leq 1.25)$

1.20 $\Rightarrow$ row 0.05 $\Rightarrow$ column wise

$$\Phi(1.25) = 0.8944$$

2) $P(Z > 1.25) = 1 - P(Z \leq 1.25)$

$$= 1 - \Phi(1.25)$$

$$= 1 - 0.8944 = 0.1056$$

3) $P(Z \leq -1.25) = \Phi(1.25)$

$$= 0.1056$$

4) $P(-0.38 \leq Z \leq 1.25)$

$$= \Phi(1.25) - \Phi(-0.38)$$

$$= 0.8944 - 0.3520 = 0.5424$$

5) $P(Z \geq 5) = \Phi(5) = 0.9998 \approx 1$

If value exceeds 3.4 we have to write 0.9998 - last value of table

[for any value greater than 3.4]

$$P(Z \leq -5) = 0.0003 \text{ [first row first value]}$$

Percentiles of Standard normal deviations

Result:

Z_α will denote the value on Z -axis for which α of the area under the Z curve lies to the right of Z_α .



Standard normal percentiles
and critical values:

Percentiles	90	95	97.5	99	99.5	99.75	99.9
α (upper-tail area)	0.1	0.05	0.025	0.01	0.005	0.001	0.0005
$Z_\alpha = 100(1 - \alpha)^{th}$ Percentile	1.28	1.645	1.96	2.33	2.58	3.08	3.27

Result: (relation btwn x & z): $x \leftrightarrow z$

① If x has normal distribution with mean μ & S.D σ then,

$$Z = \frac{x - \mu}{\sigma}$$

$$② P(a \leq x \leq b) = P\left(\frac{a - \mu}{\sigma} \leq z \leq \frac{b - \mu}{\sigma}\right)$$

$$= \Phi\left(\frac{b-\mu}{\sigma}\right) - \Phi\left(\frac{a-\mu}{\sigma}\right)$$

$$\textcircled{2} P(X \leq a) = \Phi\left(\frac{a-\mu}{\sigma}\right)$$

$$\textcircled{3} P(X \geq b) = 1 - \Phi\left(\frac{b-\mu}{\sigma}\right)$$

$$\begin{aligned} 95 &= 1.645 \\ 99 &= 2.33 \end{aligned}$$

Exp: 4.16 Pg: 162.

compute the probability that the reaction time is btwn 1.4 & 1.753. Also find the prob that the reaction time will exceed 2.5.

Given: $\mu = 1.25$
 $\sigma = 0.46$

$$\begin{aligned} \text{a) } P(1.00 \leq X \leq 1.75) &= P\left(\frac{1.00 - 1.25}{0.46} \leq Z \leq \frac{1.75 - 1.25}{0.46}\right) \\ &= P(-0.5437 \leq Z \leq 1.0869) \\ &= \Phi(1.08) - \Phi(-0.54) \\ &= 0.8599 - 0.2946 = 0.5653 \end{aligned}$$

$$\begin{aligned} \text{b) } P(X > 2) &= 1 - P(X \leq 2) \\ &= 1 - P\left(\frac{2 - 1.25}{0.46}\right) \\ &= 1 - P(Z \leq 1.63) \\ &= 1 - \Phi(1.63) \\ &= 1 - 0.9484 = 0.0516 \end{aligned}$$

Ex. 4.17

normally distributed
one st. deviation of its mean.

$$P(\mu - \sigma \leq x \leq \mu + \sigma) \quad \# \text{ converting } x \text{ to } z$$

$$P\left(\frac{\mu - \sigma - \mu}{\sigma} \leq z \leq \frac{\mu + \sigma - \mu}{\sigma}\right)$$

$$= P(-1 \leq z \leq 1)$$

$$= \Phi(1) - \Phi(-1) =$$

$$= 0.8413 - 0.1587 = 0.6826$$

Result:

$$(100p)^{\text{th}} \text{ percentile for normal } (\mu, \sigma) = \mu + \left[(100p)^{\text{th}} \text{ for standard normal} \right] \cdot \sigma$$

sp Standard normal $\rightarrow$ normal.

used: Given in x term and asked for percentile

exercice : 4.3

32, $\mu = 15$

$\sigma = 1.25$

$$a) P(X \leq 15) = P\left(\frac{15 - 15}{1.25}\right)$$

$$= P(Z \leq 0) = \Phi(0)$$

$$= 0.5$$

$$b) P(X \leq 17.5) = P\left(Z \leq \frac{17.5 - 15}{1.25}\right)$$

$$\frac{17.5 - 15}{1.25} = P(Z \leq 2) = \Phi(2)$$

$$= 0.9772$$

$$c) P(X \geq 10) = 1 - P(X \leq 10)$$

$$= 1 - P\left(Z \leq \frac{10 - 15}{1.25}\right)$$

$$= 1 - P(Z \leq -4)$$

$$= 1 - \Phi(-4) = 1 - 0.0003$$

$$d) P(14 \leq X \leq 18) =$$

$$= P\left(\frac{14 - 15}{1.25} \leq Z \leq \frac{18 - 15}{1.25}\right)$$

$$P(-0.8 \leq Z \leq 2.4) = \Phi(2.4) - \Phi(-0.8)$$

$$P(a \leq X \leq b) = \Phi(b) - \Phi(a) = 0.9918 - 0.2119$$

$$e) P(|X-15| \leq 3)$$

$$P(-3 \leq X-15 \leq 3)$$

$$P(12 \leq X \leq 18)$$

$$= P\left(\frac{12-15}{1.25} \leq Z \leq \frac{18-15}{1.25}\right)$$

$$= P(-2.4 \leq Z \leq 2.4)$$

$$= \Phi(2.4) - \Phi(-2.4)$$

$$= 0.9918 - 0.0082$$

$$= \cancel{0.9918} + 0.9836$$

$$34) \mu = 30$$

$$\sigma = 0.06$$

$$a) P(X > 0.50) = P\left(Z > \frac{0.50 - 30}{0.06}\right)$$

$$= P(Z > 3.333)$$

$$= 1 - P(Z \leq 3.333)$$

$$= 1 - \Phi(3.33)$$

$$= 1 - 0.9996$$

$$= 0.0004$$

$$b) P(X \leq 0.20)$$

$$= P\left(Z \leq \frac{0.20 - 0.30}{0.06}\right)$$

$$= P(Z \leq -1.666)$$

$$= \Phi(-1.66) = 0.0485$$

note: ———

$$\left. \begin{array}{l} 95^{\text{th}} \text{ percentile} \\ \text{for normal} \\ (\mu, \sigma) \end{array} \right\} = \mu + \left(\begin{array}{l} 95^{\text{th}} \text{ percentile} \\ \text{for standard deviation} \end{array} \right) \cdot \sigma$$

$$= 0.30 + (1.645) 0.06$$

$$= 0.3987$$

Exponential distribution

① X is said to have an exponential distribution with parameter λ ($\lambda > 0$) if the pdf of X is.

$$f(x; \lambda) = \begin{cases} \lambda e^{-\lambda x} & x \geq 0 \\ 0 & \text{otherwise.} \end{cases}$$

②. The mean of exponential distribution.

$$\mu = E(X) = \frac{1}{\lambda}$$

$$\text{Var}(X) = \sigma^2 = \frac{1}{\lambda^2}$$

$$\text{SD. } \sigma = \frac{1}{\lambda}$$

Both are same.

The exponential pdf is easily integrated to obtain cdf:

$$F(x; \lambda) = \begin{cases} 0 & x < 0 \\ 1 - e^{-\lambda x} & x \geq 0 \end{cases}$$

Exp. 4.21

$$E(x) = \frac{1}{\lambda} = 6 \quad \therefore \lambda = \frac{1}{6} = 0.1667$$

Compute the probability that

1) stress range is at most 10

2) stress range is between 5 and 10.

$$1) P(X \leq 10) = F(10; 0.1667)$$

$$F(x; \lambda) = 1 - e^{-\lambda x}$$

$$\begin{aligned} F(x; \lambda) &= 1 - e^{-(0.1667 \times 10)} \\ &= 1 - e^{-1.667} \\ &= 0.8111 \end{aligned}$$

(convert to cumulative distribution)

$$2) P(5 \leq x \leq 10)$$

$$= F(10; 0.1667) - F(5; 0.1667)$$

$$= [1 - e^{-(0.1667 \times 10)}] - [1 - e^{-(0.1667 \times 5)}]$$

$$= 0.8111 - 0.5654 = 0.2456$$

Ex: 4.4

59) $\lambda = 1$

a) expected time ; $E(x) = \frac{1}{\lambda} = 1$

b) $S.D = \frac{1}{\lambda} = 1$

c) $P(x \leq 4)$

$$F(4, 1) = 1 - e^{-4}$$

$$= 0.9816$$

d) $P(2 \leq x \leq 5)$

$$= F(5, 1) - F(2, 1) = [1 - e^{-(1 \times 5)}] - [1 - e^{-(1 \times 2)}]$$

$$= 0.9932 - 0.8646$$

$$= 0.1285$$

Module - 2

Joint Probability Distribution

Two Discrete rv

- ① The joint pmf of 2 discrete rv x & y is ~~described by~~ describes how much probability mass is placed on each possible pair of values x, y .
- ② The joint probability mass function, $p(x, y)$ is defined by,

$$P(x, y) = P(X=x \text{ and } Y=y)$$

Result:

1. $P(x, y) > 0$

2. $\sum_x \sum_y P(x, y) = 1$.

- ② The marginal probability mass function of 'x' denoted by $P_x(x) = \sum_y P(x, y)$ for each possible value of x.

③ $P_y(y) = \sum_x P(x, y)$ for each possible value of y.

19/1/99
exp: 5.1

$P(x, y)$	500	1000	5000
100	0.30	0.05	0
500	0.15	0.20	0.05
1000	0.10	0.10	0.05

- compute
- i, $P(x=100, y=500)$
 - ii, $P(x=y)$
 - iii, $P(x \geq 500)$
 - iv, P_x marginal pmf of x +
marginal pmf of y

$$P(x=100, y=500) = 0.30$$

$$\begin{aligned} P(x=y) &= P(500, 500) + P(1000, 1000) \\ &= 0.15 + 0.10 \\ &= 0.25 \end{aligned}$$

$$\begin{aligned} P(x \geq 500) &= 0.15 + 0.20 + 0.05 + \\ &\quad 0.10 + 0.10 + 0.05 \end{aligned}$$

$$P(y \geq 500) = 1 = 0.65$$

marginal pmf of x .

$$P_x(x) = \sum_y P(x, y) \rightarrow \text{for possible values of } x.$$

$$x = 100, 500, 1000 \quad (\text{Add row wise})$$

$$P_x(100) = 0.30 + 0.05 + 0 = 0.35$$

$$P_x(500) = 0.15 + 0.20 + 0.05 = 0.40$$

$$P_x(1000) = 0.25$$

$$P_x(x) = \begin{cases} 0.35 & x = 100 \\ 0.40 & x = 500 \\ 0.25 & x = 1000 \\ 0 & \text{otherwise} \end{cases}$$

Marginal pmf of y:

$$P_y(y) = \sum_x P(x, y).$$

$$P_y(500) = 0.55$$

$$P_y(1000) = 0.35$$

$$P_y(5000) = 0.10$$

$$\text{The marginal pmf of } y \text{ } P_y(y) = \begin{cases} 0.55 & y = 500 \\ 0.35 & y = 1000 \\ 0.10 & y = 5000 \\ 0 & \text{otherwise} \end{cases}$$

Two Continuous random Variable:

Let x and y are cont. rv a joint prob. density function denoted by $f(x, y)$ and defined as

$$P[(x, y) \in A] = \iint_A f(x, y) dx dy$$

$$a \leq x \leq b \quad ; \quad c \leq y \leq d$$

Note:

(i) $f(x, y) \geq 0$

$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dy dx = 1$ (legitimate pdf)

ii) marginal pdf of x and y denoted by,
 $f_x(x)$ & $f_y(y)$ and it is defined by,

$$f_x(x) = \int_{-\infty}^{\infty} f(x, y) dy \quad -\infty \leq x \leq \infty$$

$$f_y(y) = \int_{-\infty}^{\infty} f(x, y) dx \quad -\infty \leq y \leq \infty$$

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exp: 5.3.

$$f(x, y) = \begin{cases} \frac{6}{5} (x + y^2) & 0 \leq x \leq 1, 0 \leq y \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

compute i, $P(0 \leq x \leq 1/4, 0 \leq y \leq 1/4)$

ii) $P(x < 0)$

iii) P.T the function is legitimate

iv) Find the marginal pdfs x and y .

$$P(0 \leq x \leq 1/4, 0 \leq y \leq 1/4)$$

$$= \int_0^{1/4} \int_0^{1/4} \frac{6}{5} (x+y^2) dx dy \quad (2)$$

$$ii) \quad f_x(x) = \int_{-\infty}^{\infty} f(x, y) dy$$

$$f_x(x) = \int_0^1 \frac{6}{5} (x+y^2) dy$$

$$= \frac{6}{5} \int_0^1 (x+y^2) dy = \frac{6}{5} \left[xy + \frac{y^3}{3} \right]_0^1$$

$$= \frac{6}{5} \left[x + \frac{1}{3} \right] = \frac{6}{5} x + \frac{62}{155}$$

$$f_x(x) = \begin{cases} \frac{6}{5} x + \frac{2}{5} & 0 < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

$$f_y(Y) = \int_0^1 \frac{6}{5} (x + y^2) dx$$

$$= \frac{6}{5} \left[\frac{x^2}{2} + xy^2 \right]_0^1$$

$$= \frac{6}{5} \left(\frac{1}{2} + y^2 \right)$$

$$= \frac{3}{5} + \frac{6}{5} y^2$$

$$f_y(Y) = \begin{cases} \frac{3}{5} + \frac{6}{5} y^2 & 0 \leq y \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

Note:

2 r.v x and y are said to be independent if for every pair of x and y values

$$P(x, y) = P_x(x) \cdot P_y(y)$$

when x and y are discrete

$$f(x, y) = f_x(x) \cdot f_y(y)$$

when x and y are continuous

If these conditions are not satisfied

Conditional Distribution:

Let x and y be 2 rv. the joint pdf $f(x, y)$ and marginal x pdf $f_x(x)$. Then the for any x value x for which $f_x(x) > 0$.

the conditional probability density y given that $x = x$

Conditional prob density is

$$\begin{cases} f_{(y|x)}(y|x) = \frac{f(x, y)}{f_x(x)} & -\infty < y < \infty \\ f_{(x|y)}(x|y) = \frac{f(x, y)}{f_y(y)} & -\infty < x < \infty \end{cases}$$

If x and y are discrete replace pdf by pmf.

$$\begin{cases} p_{x|y}(x|y) = \frac{P(x, y)}{P_y(y)} & \text{Range of } x \\ p_{y|x}(y|x) = \frac{P(x, y)}{P_x(x)} & \text{Range of } y \end{cases}$$

↳ conditional prob. mass fn.

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ex. 5.1.

		y		
p(x,y)		0	1	2
x	0	.10	.04	.02
	1	.08	.20	.06
	2	.06	.14	.30

a, $P(X=1 \text{ and } Y=1) = 0.20$.

b, $P(X \leq 1 \text{ and } Y \leq 1)$
 $= 0.10 + 0.04 + 0.08 + 0.20$
 $= 0.42$.

c, mag pmt of x and y.

$P_X(X) = P_X(0) = 0.16$

$P_X(1) = 0.34$

$P_X(2) = 0.5$

$P_{Y,Y} = \sum_{X_i} P(X_i, Y)$

$P_{X,X} = \sum_Y P(X, Y)$

$P_X(X) = \begin{cases} 0.16 & x=0 \\ 0.34 & x=1 \\ 0.5 & x=2 \\ 0 & \text{otherwise} \end{cases}$

$$P_Y(Y) = P_Y(0) = 0.24$$

$$P_Y(1) = 0.58$$

$$P_Y(2) = 0.38$$

$$P_Y(Y) = \begin{cases} 0.24 & Y=1 \\ 0.58 & Y=2 \\ 0.38 & Y=3 \\ 0 & \text{otherwise} \end{cases}$$

$$P(X \leq 1) = 0.5$$

$$\hookrightarrow P_X(0) + P_X(1) = 0.16 + 0.34 = 0.5$$

e) are x & y independent?

$$P(x, y) = P_X(X) \cdot P_Y(Y)$$

randomly select a pair.

$$P(2, 1) = P_X(2) \cdot P_Y(1)$$

$$= 0.15 \times 0.38$$

$$= 0.19 \neq 0.14$$

$$P(0, 0) = P_X(0) \cdot P_Y(0)$$

$$= 0.16 \cdot 0.24 = 0.0384 \neq 0.10$$

	X_2			
	0	1	2	3
0	.08	.07	.04	.00
1	.06	.15	.05	.01
2	.09	.01	.10	.06
3	.00	.03	.04	.07
4	.00	.01	.05	.06

a) $P(X_1=1, X_2=1) = 0.15$

b) $P(X_1 = X_2) = 0.08 + 0.15 + 0.10 + 0.07 = 0.40$

$P(X_1=0, X_2=0) = 0.08$

$P(X_1=1, X_2=1) = 0.15$

$P(X_1=2, X_2=2) = 0.10$

$P(X_1=3, X_2=3) = 0.07$

c) total no. of customers is exactly 4?

At least four?

$\Rightarrow P(0, 3) + P(2, 2) + P(3, 1) + P(4, 0)$

$= 0.04 + 0.01 + 0.03 + 0.00 = 0.08$

(x+y)

at least 4 $P(X \geq 4)$
 $P(2, 2) + P(3, 1) + P(4, 0) + P(1, 3)$
 $P(2, 3) + P(3, 2) + P(3, 3) + P(4, 1)$
 $+ P(4, 2) + P(4, 3) = 0.46$

y/x	1	2	3	4	5	6
0	0	0	$1/32$	$2/32$	$2/32$	$3/32$
1	$1/16$	$1/16$	$1/8$	$1/8$	$1/8$	$1/8$
2	$1/32$	$1/32$	$1/64$	$1/64$	0	$2/64$

Compute

$$P(X \leq 1)$$

$$P(Y \leq 3)$$

$$P(X \leq 1, Y \leq 3)$$

$$P(X + Y \leq 4)$$

$$P(X \leq 1 | Y \leq 3)$$

$$P(\cancel{X \leq 1} | Y \leq 3)$$

$$P(\cancel{X \leq 3} | Y \leq 1) = \frac{P(X \leq 1, Y \leq 3)}{P(Y \leq 3)}$$

(2)

x/y	2	4
1	0.10	0.15
3	0.20	0.30
5	0.10	0.05

Verify X & Y
are independent
or not

2/2/25

compute the marginal density function of x and y and compute the conditional density function of y given $x=x$ for the joint Probability Function.

$$f(x, y) = \begin{cases} \frac{8}{9} xy & , 1 < x < y < 2. \\ 0 & , \text{otherwise} \end{cases}$$

Soln:

$$F_x(x) = \int_{-\infty}^{\infty} f(x, y) dy.$$

$$= \int_x^2 \frac{8}{9} xy dy.$$

$$= \left[\frac{8}{9} \frac{xy^2}{2} \right]_x^2 = \frac{16x}{9} - \frac{4x^3}{9}$$

$$= \frac{4x}{9} (4x - x^3) ; 1 < x < 2.$$

$$F_y(y) = \int_{-\infty}^{\infty} f(x, y) dx.$$

$$= \int_1^y \frac{8}{9} xy dx = \left[\frac{8}{9} \frac{x^2 y}{2} \right]_1^y$$

$$= \frac{4}{9} (y^3 - y) \quad ; \quad 1 < y < 2.$$

$$f(y|x)^{(1/x)} = \frac{f(x,y)}{f_x(x)} = \frac{\frac{28}{9} xy}{\frac{4}{9} (4x - x^3)}$$

$$= \frac{2xy}{4x - x^3} = \frac{2y}{4 - x^2} \quad (1 < x < y < 2)$$

2. The joint density function $f(x,y) e^{-(x+y)}$

$0 \leq x, y \leq \infty$, verify whether x & y are independent.

$$f(x,y) = f_x(x) \cdot f_y(y).$$

$$f_x(x) = \int_{-\infty}^{\infty} e^{-(x+y)} dy$$

$$= \int_0^{\infty} e^{-(x+y)} dy = \int_0^{\infty} (e^{-x} \cdot e^{-y}) dy$$

$$= e^{-x} \int_0^{\infty} e^{-y} dy = e^{-x} [-e^{-y}]_0^{\infty}$$

$$= ? \quad = e^{-x}$$

$$f_y(y) = \int_0^{\infty} e^{-(x+y)} dx$$

$$= e^{-y} \int_0^{\infty} e^{-x} dx$$

$$= e^{-y}$$

$$f_x(x) \cdot f_y(y) = e^{-x} e^{-y} = e^{-(x+y)} = f(x,y)$$

So Independent.

Q, The Joint pdf. is given by

$$f(x,y) = k \cdot x^2(8-y), \quad x < y < 2x, \quad 0 < x < 2$$

compute 'k'

Soln:

w.k.t.

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x,y) dy dx = 1.$$

$$\int_0^2 \int_x^{2x} (k x^2 (8-y)) dy dx = 1$$

$$\int_0^2 \int_x^{2x} (8kx^2 - kx^2y) dy dx = 1$$

$$\int_0^2 \left[8kx^2y - \frac{k y^2 x^2}{2} \right]_x^{2x} dx = 1$$

$$= \left[16x^3k - \frac{4x^4k}{2} \right]_0^2 = 1$$

$$16kx^3 - 8kx^3 - \frac{8kx}{2} + \frac{kx^4}{2} = 1$$

$$\int_0^2 8kx^3 - kx^4 + \frac{kx^4}{2} dx = 1$$

$$\int_0^2 \frac{16kx^3 - 2kx^4 + kx^4}{2} dx = 1$$

$$\frac{1}{2} \int_0^2 16kx^3 - 2kx^4 + kx^4 dx = 1$$

$$\frac{1}{2} \left[\frac{16kx^4}{4} - \frac{kx^5}{5} \right]_0^2 = 1$$

$$\frac{1}{2} \left[4kx^4 - \frac{kx^5}{5} \right]_0^2 = 1$$

$$\frac{1}{2} \left[64k - \frac{32k}{5} \right] = 1$$

$$\frac{1}{2} \left[\frac{320k - 32k}{5} \right] = 1 \Rightarrow \frac{288k}{5} = 1$$

$$k = \frac{5}{288}$$

$$\boxed{\text{Ans} = 0.446}$$

5) The joint pdf is

$$f(x, y) = \begin{cases} \frac{b-x-y}{8} & ; 0 < x < 2, 2 < y < 4 \\ 0 & ; \text{otherwise.} \end{cases}$$

compute $f(y|x=2)$

$$F_{y|x=2} (y|x=2) = ?$$

$$f(y|x=2) = \frac{f(x, y)}{f_x(x=2)} \quad (\text{or}) \quad \frac{f(x, y)}{f_x(x)}.$$

$$f_x(x) = \int_2^4 \frac{b-x-y}{8} dy = \frac{1}{8} \left[by - xy - \frac{y^2}{2} \right]_2^4$$

$$= \frac{1}{8} [b-2x]$$

$$f_x(x=2) = \frac{2}{8} = \frac{1}{4}$$

$$f(y|x=2) = \frac{\frac{b-x-y}{8}}{1/4} = \frac{b-2-y}{8}$$

$$= \frac{4-y}{8} \quad \xrightarrow{1/4} = \frac{4-y}{2}$$

24th feb

2/05/25

Definition:

Expected value, Covariance & Correlation

Let x & y be jointly distributed rv's with pmf $p(x, y)$ (or) pdf $f(x, y)$ according to whether the variables are discrete or continuous. Then the expected value of function $h(x, y)$.

$$E[h(x, y)] = \begin{cases} \sum_x \sum_y h(x, y) \cdot p(x, y) & \Rightarrow \text{discrete} \\ \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} h(x, y) f(x, y) dx dy & \Rightarrow \text{continuous} \end{cases}$$

$$E(x) = \sum \sum p(x, y) \cdot x$$

$$E(xy) = \sum \sum p(x, y) xy$$

$$E(xy) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) xy$$

The covariance of 2 rv's x & y is

$$\text{Cov}(x, y) = E(xy) - E(x)E(y)$$

Correlation coefficient of x & y is.

$$\text{corr}(x, y) = \rho_{xy} = \frac{\text{cov}(x, y)}{\sigma_x \sigma_y}$$

19 m

$$\sigma_x = \sqrt{\text{Var}(x)}$$

$$\text{Var}(x) = E(x^2) - [E(x)]^2$$

$$\boxed{-1 \leq \text{corr} \leq 1}$$

① 2 r.v's x & y have the following joint pdf

$$f(x, y) = \begin{cases} 2-x-y, & 0 \leq x \leq 1, 0 \leq y \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

compute correlation of x & y

$$\text{corr}(x, y) = ?$$

Soln:

$$f_x(x) = \int_0^1 f(x, y) dy$$

$$= \int_0^1 (2-x-y) dy$$

$$= \left[2y - xy - \frac{y^2}{2} \right]_0^1$$

$$= \left[\left(2 - x - \frac{1}{2} \right) - 0 \right]$$

$$= 2 - 1.5 - x \quad \text{corr} \quad \frac{3}{2} - x$$

12 to 15 m

$$f_y(y) = \int_0^1 f(x,y) dx$$

$$= \int_0^1 (2-x-y) dx = \left[2x - \frac{x^2}{2} - xy \right]_0^1$$

$$= \frac{3}{2} - y$$

$$E(x) = \int_0^1 x \cdot f_x(x) dx$$

$$= \int_0^1 x \left(\frac{3}{2} - x \right) dx = \int_0^1 \left(\frac{3}{2}x - x^2 \right) dx$$

$$= \left[\frac{3x^2}{4} - \frac{x^3}{3} \right]_0^1 = \frac{3}{4} - \frac{1}{3} = \frac{5}{12}$$

$$E(x^2) = \int_0^1 x^2 f_x(x) dx$$

$$= \int_0^1 x^2 \left(\frac{3}{2} - x \right) dx$$

$$= \int_0^1 x^2 \left(\frac{3}{2} x^2 - x^3 \right) dx$$

~~$$\left[\frac{3x^5}{10} - \frac{x^6}{6} \right]_0^1 = \frac{3}{10} - \frac{1}{6} = \frac{1}{5}$$~~

$$= \int_0^1 x^2 \left(\frac{3}{2} - x \right) dx = \int_0^1 \frac{3}{2} x^2 - x^3 dx$$

$$= \left[\frac{3x^3}{6} - \frac{x^4}{4} \right]_0^1 = \frac{1}{2} - \frac{1}{4} = \frac{1}{4}$$

$$E(Y) = \int_0^1 y \cdot f_Y(y) dy$$

$$= \int_0^1 y \left(\frac{3}{2} - y \right) dy = \int_0^1 \frac{3}{2} y - y^2 dy$$

$$= \left[\frac{3}{4} y^2 - \frac{y^3}{3} \right]_0^1 = \frac{3}{4} - \frac{1}{3} = \frac{5}{12}$$

$$E(Y^2) = \int_0^1 y^2 f_Y(y) dy.$$

$$= \int_0^1 y^2 \left(\frac{3}{2} - y \right) dy.$$

$$= \int_0^1 \left(\frac{3}{2} y^2 - y^3 \right) dy = \left[\frac{3y^3}{6 \cdot 2} - \frac{y^4}{4} \right]_0^1$$

$$= \frac{1}{2} - \frac{1}{4} = \frac{1}{4}.$$

$$\text{Var}(X) = E(X^2) - [E(X)]^2$$

$$\text{Var}(Y) = E(Y^2) - [E(Y)]^2$$

$$\text{Var}(X) = \frac{5}{12} - \frac{1}{4} - \left(\frac{5}{12} \right)^2 = \frac{11}{144}$$

$$\text{Var}(Y) = \frac{1}{4} - \left(\frac{5}{12} \right)^2 = \frac{11}{144}$$

$$\sigma_x = \sqrt{\text{Var}(x)} = \sqrt{0.2762} = 0.5256$$

$$\sigma_y = \sqrt{\text{Var}(y)} = \sqrt{0.2762} = 0.5256$$

$$E(xy) = \int_0^1 \int_0^1 xy \cdot f(x,y) \, dx \, dy$$

$$= \int_0^1 \int_0^1 xy \cdot (2-x-y) \, dx \, dy$$

$$= \int_0^1 \int_0^1 (2xy - x^2y - xy^2) \, dx \, dy$$

$$= \int_0^1 \left[\frac{2x^2y}{2} - \frac{x^3y}{3} - \frac{xy^2}{2} \right]_0^1 \, dy$$

$$= \frac{1}{6}.$$

$$\text{Cov}(x, y) = E(xy) - E(x)E(y)$$

$$= \frac{1}{6} - \left(\frac{5}{12}\right)\left(\frac{5}{12}\right)$$

$$= -\frac{1}{144}.$$

$$= -0.0069.$$

$$\text{Corr}(x, y) = \frac{\text{Cov}(x, y)}{\sigma_x \sigma_y}.$$

$$= \frac{-0.0069}{(0.2762)(0.2762)}.$$

$$\rho_{xy} = -0.0904.$$

Practice problems:

The joint pdf is given by.

$$f(x, y) = \begin{cases} \frac{3}{2} (x^2 + y^2), & 0 \leq x \leq 1, \\ & 0 \leq y \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

compute ρ_{xy} .

② The joint distribution of x and y is given

by $f(x, y) = \frac{x+y}{21}$ $x = 1, 2, 3$ compute P_{xy}
 $y = 1, 2$.

Soln.

This is discrete type

① go to table.

then $P_x x$, $P_y y$, $E(x)$, $E(y)$...

~~marginal~~

How to put table?

C>

y/x	1	2	3	$P_y(y)$
1	$2/21$	$3/21$	$4/21$	$9/21$
2	$3/21$	$4/21$	$5/21$	$12/21$
$P_x(x)$	$5/21$	$7/21$	$9/21$	

$$P(x=1) = 5/21$$

$$P(x=2) = 7/21$$

$$P(x=3) = 9/21$$

$$P(y=1) = 9/21$$

$$P(y=2) = 12/21$$

$$P_x(x) = \begin{cases} 5/21 & , x=1 \\ 7/21 & , x=2 \\ 9/21 & , x=3 \\ 0 & , \text{otherwise} \end{cases}$$

$$P_y(y) = \begin{cases} 9/21 & , y=1 \\ 12/21 & , y=2 \\ 0 & , \text{otherwise} \end{cases}$$

$$E_x(x) = \sum x \cdot P_x(x)$$

$$E_y(y) = \sum y \cdot P_y(y)$$

Module - 3

Test of Hypotheses Based on a Single Sample.

3 different test.

- 1, z test # sample is big.
 - 2, T test # " " small
 - 3, Population proportion # data in form of proportion
- (Previous book)

when $n > 30 \rightarrow$ large sample. (Pract edition)

Remarks

- ① A statistical hypothesis is a claim (or) assertion ^{H_0} either about the ^{H_a} (value of a single parameter)
- ② NULL HYPOTHESIS, denoted by H_0 , is the claim that 'It is initially assumed to be true'
- ③ ALTERNATE HYPOTHESIS, denoted by H_a , is the assertion that is 'contradictory to H_0 '

eg:

$$H_0 : \mu = \mu_0$$

$$H_a = \mu \neq \mu_0$$

($\mu \downarrow$ left, $\mu \uparrow$ right) $\mu < \mu_0$ $\mu > \mu_0$

Two tailed test

left tailed test

right tailed test

④ A Test of Hypothesis is a method for using sample data to decide whether the null hypothesis ^{should} to be rejected.

⑤ Test Statistic is the function of a sample data on which the decision is to be based.

⑥ A rejection region is ^{the} set of all test statistic value for which H_0 will be rejected.

⑦ Type I error consists of (rejecting) the null hypothesis H_0 , when it is true.

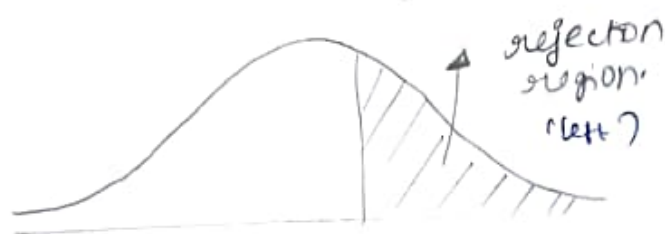
⑧ Type II error consists of (not rejecting) H_0 , when H_0 is false.

⑧

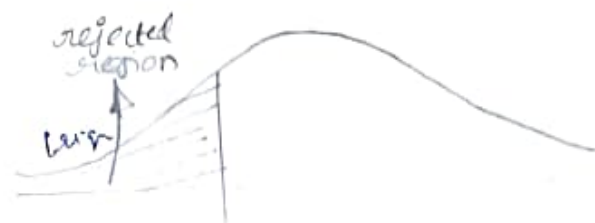
Two tailed test :



Right Tailed test :



Left tailed test :



9, The resulting value of α is referred to the significance level of the test.

The traditional levels of significance are,

90 - 10 = 100	*	0.10	→ 90%
$\frac{10}{100} = 0.10$	*	0.05	→ 95% (most preferred)
	*	0.01	→ 99%

$$\alpha = 0.05$$

⑩ Defn of p value. /

Pvalue - p value is the probability calculated assuming the null hypothesis is true. A conclusion is reached in a hypothesis by selecting number α of the test, that is reasonably close to zero. H_0 will be

rejected if $\boxed{p\text{value} \leq \alpha}$ (✓)
 H_0 will be accepted if $\boxed{p\text{value} > \alpha}$
 (or)
 fail to reject

Test about a population mean - Z test.

mean, variance, pop. proportion.

parameter; μ

There are ^{six} ~~seven~~ steps to be followed:

steps for testing hypothesis:

- 1, Identify the parameter (μ).
- 2, Determine the null value & alternate hypothesis (H_0).
- 3, Give the formula for the computed value of test statistics (Z).
- 4, State the rejection region for significance value α .
- 5, Compute the test statistic value - (Z or t).
- 6, Decide whether H_0 should be rejected
(get value of P)
If α provided take that else $\boxed{\alpha = 0.05}$
get value from table.

- 1, Identify the parameter (μ)
- 2, Determine the null value & alternate hypothesis (H_0)

3, Give the formula for the computed value of test statistics (Z)

4. State the rejection region for significance value α -

5. Compute the test statistic value - $(Z \text{ or } t)$.
Whether H_0 ~~is~~ should be rejected

5, Compute the test statistic value

6, Decide whether H_0 ~~is~~ should be rejected
(get value of P)
i.e. take that else $\alpha = 0.05$

If α provided take that else $\alpha = 0.05$

Pvalue $\rightarrow$ probability value from table.

$P\text{-value} > \text{Test Statistic value}$ [accept H_0]

$p\text{-value} > \text{static value}$
 $p\text{-value} \leq \text{static value}$ [reject H_0]

Formulae;

Z test:

Null hypothesis $H_0: \mu = \mu_0$

Alternative hypothesis $H_a: \mu \neq \mu_0$

$H_a: \mu > \mu_0$

$H_a: \mu < \mu_0$

P value

2 times area under the normal curve to the right of $|z|$
area under the normal curve to the right of z
area under the normal curve to the left of z

$H_a: \mu \neq \mu_0 = 2(1 - \phi(z)) \rightarrow$ two tailed test

$H_a: \mu > \mu_0 = 1 - \phi(z)$

$H_a: \mu < \mu_0 = \phi(z)$

make z value positive
else have it as -ve

n - sample size

s - S.D of sample

σ - S.D of population

$\bar{x}$ - sample average

μ - population average

μ_0 - population mean

Example: 8.6 pg: 329

~~Soln:~~

Collect the parameter.

Parameter, μ

Given; $n = 9$

Example 36.

Solution:

Parameter is ' μ ' [average]
↳ word is repeated.

Given: sample size, $n = 9$.

pop. average, $\mu = 130^\circ$

sample average, $\bar{X} = 130.08^\circ$

S.P.

$\sigma/\sqrt{n}$

pop. S.D., $\sigma = 1.5$

$\alpha = 0.01$.

5) Define null hypothesis, $H_0: \mu = \mu_0$.

Alternative hypothesis, $H_a: \mu \neq \mu_0$.

↳ contradict.

(choose $\neq, >, <$ from the question - read qn)

6) write formula for Z , $Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$

$$7) Z = \frac{131.08 - 130}{1.5 / \sqrt{9}} = 2.19.$$

$$\text{P-value, } p = 2(1 - \phi(|Z|))$$

$$= 2(1 - \phi(2.16))$$

$$= 2(1 - 0.9846)$$

$$= 0.0308.$$

$$p\text{-value} > \alpha\text{-value. (con.)}$$

$$0.0308 > 0.01$$

Conclusion: Accept H_0 . The data does not contradict.

Ex: 8.2

18, Test the statistics.

Given, $\bar{x} = 72.3$.

$$\sigma = 9.$$

$$H_0: \mu = 75.$$

$$\alpha = 0.02.$$

$$H_a: \mu < 75.$$

$$n = 25.$$

$$Z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}} = \frac{72.3 - 75}{9 / \sqrt{25}} = -1.5$$

$$p\text{-value} = \phi(-1.5).$$

$$= 0.0668$$

$$p\text{value} > \alpha\text{value.}$$

$$0.0668 > 0.02.$$

Accept H_0 .

19)

Given;

$$\bar{x} = 94.32.$$

$$n = 16.$$

$$\sigma = 1.20.$$

$$H_0: \mu = 95.$$

$$H_a: \mu \neq 95.$$

$$\alpha = 0.01.$$

$$Z = \frac{\bar{X} - \mu}{\sigma / \sqrt{n}} = \frac{94.32 - 95}{1.20 / \sqrt{16}}$$

$$Z = -2.2666.$$

$$p\text{value}, p = 2(1 - \phi(|Z|)).$$

$$= 2(1 - \phi(2.2666))$$

$$= 2(1 - 0.9881)$$

$$= 0.0238$$

$$p\text{value} > \alpha\text{value}.$$

$$0.0238 > 0.01.$$

Accept H_0 .

$$21, \quad n=16 \quad \mu = 5.5$$

$$\sigma = 0.3$$

$$\bar{x} = 5.25$$

$$Z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} = \frac{5.25 - 5.5}{0.3/\sqrt{16}} = -3.33$$

$$H_0: \mu = 5.5 \quad \rightarrow \text{[disturbs from true average]}$$

$$H_a: \mu \neq 5.5 = 2(1 - \phi(1-3.33))$$

$$= 2(1 - \phi(3.33))$$

$$= 2(1 - 0.9996)$$

$$= 0.0008$$

$$\alpha = 0.05$$

$$p < 0.05$$

reject H_0 .

$$22, \quad n=25$$

$$\bar{x} = 52.7$$

$$\mu = 50$$

$$\sigma = 4.8$$

$$H_0: \mu = 50$$

$$H_a: \mu \leq 50 =$$

exp 8.8

18/02/24.

true average:

$$H_0: \mu = 30$$

$$H_a: \mu < 30 \rightarrow \text{left sided.}$$

$$n = 52.$$

Find $\bar{x}$ & σ in the sample.

$$\mu = 30.$$

$$\bar{x} = 28.7615$$

$$s = 12.26 \Rightarrow \text{Here given a sample so we take}$$

$$p\text{-value} = P(\Phi(Z)). \quad s\text{-sample SD}$$

$$Z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} = -0.7281.$$

$$P \Rightarrow \Phi(-0.7281) = 0.2358.$$

$$\text{let } \alpha = 0.05.$$

$$p > \alpha$$

Obs: Accept H_0 .

Calculator

mode
shift + mode.
↳ Reset all ==
press mode 2 times
press 1 SD.
First number
then press M+

press AC button
shift + 2.

press 1,
mean is round.

$\bar{x}, s$

press AC.

shift + 2.

$\sigma_n \rightarrow$ pop. SD deviation
option 3 - press.

$\sigma_{n-1} \rightarrow$ sample SD.

$\bar{x} \Rightarrow 1$ press

$s = 3$ press.

enter sum - reset.

8.2

17) $H_0: \mu = 30,000.$

$H_a: \mu > 30,000$

$n = 16$ $\sigma = 500$

$\alpha = 0.01$

$\bar{x} = 30,960.$

$$Z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}} \quad Z = 2.56.$$

$$\begin{aligned} p &= 1 - \phi(Z) = 1 - \phi(2.56) \\ &= 1 - 0.9948 \\ &= 0.0052 \end{aligned}$$

$p\text{-value} < \alpha\text{-value}$

Conclusion: reject H_0 .

$$25) \quad n = 52.$$

$$\bar{x} = 1.95.$$

$$s = 0.20$$

$$H_0: \mu = 2.$$

$$\mu = 2$$

$$H_a: \mu < 2. \rightarrow \text{left} = \phi(z).$$

$$\alpha = 0.01;$$

$$z = \frac{\bar{x} - \mu}{s/\sqrt{n}} = \frac{1.95 - 2}{0.20/\sqrt{52}}.$$

$$= -1.8027.$$

$$p = \phi(-1.8027).$$

$$= 0.0359.$$

$$p\text{value} > \alpha\text{value}.$$

Accept H_0 .

Solve problem: 22.

One Sample T-Test:

Null

1. Parameter is μ .

2. Null hypothesis

$$H_0: \mu = \mu_0.$$

Alternative hypothesis

$$H_a: \mu \neq \mu_0$$

$$2 \phi (1)$$

$$\mu < \mu_0 \rightarrow 1 - \phi (t)$$

$$\mu > \mu_0 \rightarrow \phi (t)$$

~~$H_a: \mu \neq \mu_0$~~

notation is imp.

Test statistic, $t = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$

Test is based on degrees of freedom:

$$\boxed{\gamma = n - 1}$$

Pg: 337.

exp: 8, 9.

$$H_a: \mu = 500$$

$$H_a: \mu > 500$$

$$n = 19.$$

$$\alpha = 0.05$$

$$\bar{x} = 562.08$$

$$s = 180.874$$

$$t = \frac{\bar{x} - \mu}{s / \sqrt{n}} = 1.51$$

$\gamma^2 = n - 1 = 19 - 1 = 18$ [select t & γ^2 in t-table]

degrees of freedom = 18

p-value: $\phi(1.51) = 0.075$

p-value > α -value

0.075 > 0.05

pg: 344

29, i, $n = 13$

$t = 1.6$

$\alpha = 0.05$

μ

$\bar{x}$

s or σ

$H_a: \mu \neq 0.5$

$H_0: \mu = 0.5$

$\gamma = n - 1 = 12$

p-value = $2 \phi(t)$

= $2 \phi(1.6)$

= 0.1336

p-value > α -value

$$31, a, n=15 \quad t=3.2 \quad \alpha=0.05.$$

$$df = n-1 = 14.$$

$$H_0: \mu = 20$$

$$H_a: \mu > 20.$$

$$P\text{-value} : \phi(3.2).$$

$$= 0.003 < \alpha \text{ value.}$$

Reject H_0 .

$$30, a, n=6$$

$$\bar{x} = 6.68.$$

$$s/\sqrt{n} = 0.0820.$$

$$H_0: \mu = 7.$$

$$H_a: \mu < 7. = 1 - \phi(t)$$

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}} = \frac{6.68 - 7}{0.0820} = -3.902$$

-ve t table is not there.

$$\text{so, } 1 - (1 - \phi(t)) \checkmark$$

$$1 - [1 - \phi(t)]$$

$$1 - 1 + \phi(t)$$

$$\Rightarrow \phi(t)$$

Note: ϕ -value +ve means - Left tailed.

$$\Rightarrow 1 - \phi(t).$$

$$t \text{ -value -ve} \Rightarrow \phi(t).$$

p-value
 Left tailed. $= \Phi(3.902)$
 $= 0.006$

p-value $<$ α -value.

Reject H_0

03/03/25

Cat-2

Testing Tests concerning a population proportion

Let p denote the proportion of
 individuals or objects in a population
 who possess a specified property.
 If an individual or object with a property
 is labeled as success 's', then (p) is the
population proportion of success.

Null hypothesis $H_0 : p = p_0$

Test statistics value $Z = \hat{p} - p_0$

$$\sqrt{p_0(1-p_0)/n}$$

$\hat{p} \rightarrow$ sample proportion $= \frac{x}{n}$

Alternative